

Energy Trade Operations

Excel Portfolio Preview

Selected excerpts from a fictional Houston physical-energy workbook - Version 25

Trade Settlements | Middle Office | Operations Controls | Excel + Python

Portfolio disclaimer

All data, entities, amounts and scenarios are fictional or synthetic. This preview contains no employer, customer or counterparty confidential data.

12	23	44	105	0	PASS
Physical trades	Invoices	Worksheets	Structured tables	Formula errors	Workbook QA

	A	B	C	D	E	F	G	H	I	J	K
1											
2											
3											
4	Project Purpose										
5	The workbook models a Houston-based physical energy trading operation across crude oil, refined products, fuel oil and NGLs. It is designed to demonstrate how commercial terms and physical events create operational, accounting and settlement consequences. The control architecture uses normalized IDs, master-data rules, formula-driven reconciliation and Python-ready tables.							Metric		Value	
6								Physical Trades		12	
7								Movements		12	
8								Locations / Terminals		13	
9								Invoices		23	
10								Open Claims		5	
11								Operational Exceptions		31	
12	Operations-to-Settlement Lifecycle							Closing Inventory Value		\$95,603,902	
13											
14											
15								Material PLA Reviews		8	
16											

Workbook scope spans the lifecycle from trade capture and physical execution through invoicing, payments, cash forecasting and month-end close.

1. Executive Control Dashboard

Workbook sheet: Dashboard

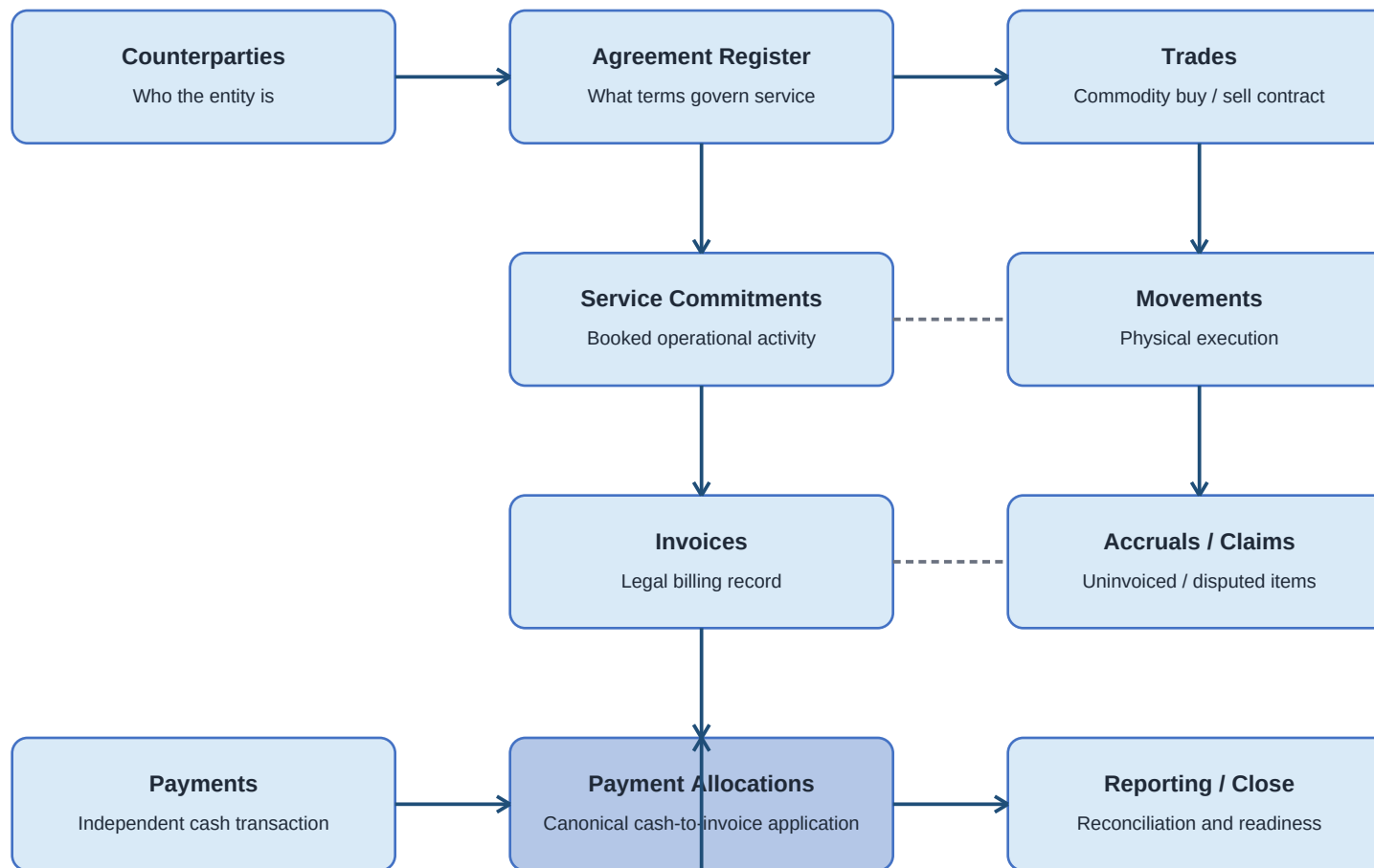
	A	B	C	D	E	F	G	H	I	J	K	L
1	Houston Gulf Coast Energy Trade Operations Demurrage & Reconciliation Dashboard											
2	Expanded 12-trade fictional portfolio covering commercial terms, physical execution, master-data readiness, inventory, service costs, claims, invoices, payments, cash forecasting and month-end close.											
3												
4												
5												
6												
7	Operations KPI		Value									
8	Total Trades		12									
9	Open Movements		0									
10	Delayed Movements		4									
11	Missing Documents		3									
12	Open Exceptions		31									
13	Demurrage Exposure		\$90,000.00									
14	Invoice Mismatches		4									
15	Unpaid Invoices		14									
16	High Priority Escalations		10									
17	Inventory Reconciliation Exceptions		0									
18	Incoterm Allocation Exceptions		0									
19	Service Fee Cost Exceptions		1									
20	Location Master Data Exceptions		0									
21	Duplicate Invoice Rows		2									
22	Material PLA Loss Reviews		8									
23	Total Excess PLA Loss Exposure		\$233,821.00									
24	Credit Limit Breaches		1									
25	AR Over Credit Limit		\$1,096,600.00									
26	Counterparty Readiness Exceptions		1									
27	Counterparty Master Data Exceptions		0									
28	Open Accrual Vendor Reviews		1									

What this demonstrates

- Executive monitoring across operations, documents, demurrage, invoices, credit, accruals and close.
- Month-end 'NOT READY' is an intentional operating state driven by open items - not a spreadsheet defect.
- Control KPIs surface outstanding receivables/payables, missing documents, open claims and escalation volumes.

2. Relational Architecture

The workbook separates master data, commercial terms, physical execution, legal billing, cash transactions and application logic. Payment Allocations is the canonical bridge between cash and invoices.



Design principle

A fully allocated payment does not automatically mean an invoice is fully settled. Cash evidence, allocation evidence and invoice balance evidence remain separate control concepts.

3. Cash Application and Payment Allocation

Workbook sheet: Payment Allocations

	A	B	C	D	E	F	G	H	I	J	K	L	M
1													
2													
3													
4	Allocation ID	Payment ID	Invoice ID	Trade ID	Direction	Currency	Applied Date	Applied Amount	Payment Amount	Payment Allocated Total	Payment Unallocated	Allocation Status	Allocation ID Control
5	PA-001	PAY-3001	INV-2001	TRD-1001	Outbound	USD	2026-08-27	\$15,261,015.00	\$15,261,015.00	\$15,261,015.00		- PASS	PASS
6	PA-002	PAY-3002	INV-2003	TRD-1003	Inbound	USD	2026-08-24	\$20,090,600.00	\$20,090,600.00	\$20,090,600.00		- PASS	PASS
7	PA-003	PAY-3003	INV-2005	TRD-1005	Inbound	USD	2026-08-26	\$4,000,000.00	\$8,781,192.00	\$4,000,000.00	\$4,781,192.00	UNALLOCATED	PASS
8	PA-004	PAY-3004	INV-2006	TRD-1006	Outbound	USD	2026-08-22	\$5,232,444.00	\$5,232,444.00	\$5,232,444.00		- PASS	PASS
9	PA-005	PAY-3005	INV-2102	TRD-1003	Outbound	USD	2026-08-27	\$686,125.00	\$686,125.00	\$686,125.00		- PASS	PASS
10	PA-006	PAY-3006	INV-2104	TRD-1003	Outbound	USD	2026-08-28	\$22,750.00	\$22,750.00	\$22,750.00		- PASS	PASS
11	PA-007	PAY-3007	INV-2201A	TRD-1007	Inbound	USD	2026-08-31	\$8,691,648.00	\$8,691,648.00	\$8,691,648.00		- PASS	PASS
12	PA-008	PAY-3008	INV-2201B	TRD-1007	Inbound	USD	2026-09-05	\$3,794,432.00	\$3,794,432.00	\$3,794,432.00		- PASS	PASS
13	PA-009	PAY-3009	INV-2213	TRD-1007	Outbound	USD	2026-09-02	\$510,150.00	\$510,150.00	\$510,150.00		- PASS	PASS
14	PA-010	PAY-3010	INV-2202	TRD-1008	Outbound	USD	2026-09-10	\$7,343,280.00	\$7,343,280.00	\$7,343,280.00		- PASS	PASS
15	PA-011	PAY-3011	INV-2204	TRD-1010	Inbound	USD	2026-09-12	\$9,876,200.00	\$9,876,200.00	\$9,876,200.00		- PASS	PASS

Core control

Payment Amount - Allocated Total = Unallocated Cash

PAY-3003 illustrates a partial application: \$8,781,192 cash received, \$4,000,000 applied, leaving \$4,781,192 unallocated.

Control architecture

Payments remain independent cash transactions. Payment Allocations records the authoritative application trail and supports one-to-many and many-to-one settlement relationships.

4. Invoice Balance Re-performance

		W	X	Y	Z	AA	AB	AC	AD
2									
3		Invoice Total	Allocated Amount	Outstanding Amount	Calculated Outstanding Amount	Settlement Position	Reported Outstanding Amount	Outstanding Variance	Balance Reconciliation Status
4		\$15,261,015.00	\$15,261,015.00	\$0.00	\$0.00	SETTLED	\$0.00	\$0.00	PASS
5		\$11,071,600.00	\$0.00	\$11,071,600.00	\$11,071,600.00	OPEN BALANCE	\$11,071,600.00	\$0.00	PASS
6		\$20,090,600.00	\$20,090,600.00	\$0.00	\$0.00	SETTLED	\$0.00	\$0.00	PASS
7		\$49,241,075.00	\$0.00	\$49,241,075.00	\$49,241,075.00	OPEN BALANCE	\$49,241,075.00	\$0.00	PASS
8		\$8,781,192.00	\$4,000,000.00	\$4,781,192.00	\$4,781,192.00	OPEN BALANCE	\$0.00	\$4,781,192.00	REVIEW
9		\$5,232,444.00	\$5,232,444.00	\$0.00	\$0.00	SETTLED	\$0.00	\$0.00	PASS
10		\$409,180.00	\$0.00	\$409,180.00	\$409,180.00	OPEN BALANCE	\$409,180.00	\$0.00	PASS

Two separate questions

Invoice Total - Allocated Amount = Calculated Outstanding

Calculated Outstanding - Reported Outstanding = Variance

Settlement Position describes the economic state (SETTLED / OPEN BALANCE / OVERPAID). Balance Reconciliation Status asks whether the source-reported balance agrees with the re-performed calculation.

	AV	AW	AX
2		Invoice Balance Reconciliation	
3	Row Labels	Count of Invoice ID	Sum of Outstanding Variance
4	CONTROLLED EXCLUSION	1	\$1,542,870.00
5	REVIEW	2	\$6,781,192.00
6	Grand Total	3	\$8,324,062.00
7			

Invoice	Calc. Outst.	Position	Reported	Variance	Recon / Scope
INV-2005	\$4,781,192	OPEN BALANCE	\$0	\$4,781,192	REVIEW
INV-2201B	\$2,000,000	OPEN BALANCE	\$0	\$2,000,000	REVIEW
INV-2206D	\$1,542,870	CONTROLLED EXCLUSION	\$0	\$1,542,870	CONTROLLED EXCLUSION

Reported Outstanding Amount is a source-system balance snapshot; Calculated Outstanding is independently re-performed from invoice total and allocation detail.

5. Exception Management and Operational Review

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	
1	Compatibility & Feasibility Control Summary									Settlement & Reconciliation Summary					
2															
3	Trade Checks		Count	Movement Checks		Count	Incoterm Control		Count	Settlement Status		Count	Operational Metrics		Value
4	PASS		12	PASS		11	PASS		12	SETTLED		3	Outstanding Receivables		\$22,955,844.00
5	REVIEW		0	REVIEW		1	REVIEW		0	PENDING PAYMENT		5	Outstanding Payables		\$50,736,749.00
6	FAIL		0	FAIL		0	FAIL		0	HOLD		4	Missing Requirements		3
7													Open Claims		5
8	Control meanings											Service Fee Control		Count	
9	PASS		Commercial, asset, route, location, Incoterm and service-cost combinations are supported.									PASS	3		
10	REVIEW		A conditional operating rule, incomplete document, fee variance, duplicate invoice, approved override or physical loss above PLA requires review.									REVIEW	1		
11	FAIL		A structural relationship, location readiness, inventory balance or incompatible operating rule has failed.									FAIL	0		
12															
13	Current Operational Exceptions											PLA Loss Control		Count	Exposure
14	Exception ID	Trade ID	Movement ID	Category	Severity	Owner	Status	Exception Amount	Source Record	Description					
15	EXC-1001	TRD-1002	MOV-1002	Outstanding Receivable	High	M. Rossi	Open	\$11,071,600.00	INV-2002	Commodity invoice remains unpaid.			Within PLA	0	
16	EXC-1002	TRD-1002	MOV-1002	Demurrage	Medium	M. Rossi	Submitted	\$25,000.00	CLM-5001	Demurrage claim submitted for 24 excess laytime hours.			Above PLA	12	
17	EXC-1003	TRD-1004	MOV-1004	Document / Price Match	High	M. Rossi	On Hold	\$29,825.00	INV-2004	Signed SOF is missing and the supplier invoice price exceeds the contract price by \$0.05/bbl.			Material Open	8	
18	EXC-1004	TRD-1004	MOV-1004	Demurrage	Medium	M. Rossi	Draft	\$35,000.00	CLM-5002	Demurrage calculation is complete, but supporting documentation is incomplete.			Total Excess Exposure		\$233,821.00
19	EXC-1005	TRD-1005	MOV-1005	Quantity / Document / Payment	High	S. Smith	On Hold	\$4,781,192.00	INV-2005	Invoice quantity exceeds delivered quantity, final COA is missing and only a partial payment has been received.					
20	EXC-1006	TRD-1001	MOV-1001	Freight Payable	Medium	S. Smith	Open	\$409,180.00	INV-2101	Approved ocean-freight invoice remains unpaid.			Credit PASS	19	
21	EXC-1007	TRD-1005	MOV-1005	Freight Payable	Medium	S. Smith	Open	\$57,456.00	INV-2103	Approved barge-freight invoice remains unpaid.			Credit REVIEW	0	
22	EXC-1008	TRD-1002	MOV-1002	Demurrage Receivable	Medium	M. Rossi	Open	\$25,000.00	INV-2105	Approved demurrage invoice has been issued and remains unpaid.			Credit FAIL	1	
23	EXC-1009	TRD-1002	MOV-1002	Uninvoiced Freight Accrual	High	M. Rossi	Aged Open	\$686,125.00	ACC-1004	Completed HSFO voyage has an aged freight accrual and no carrier invoice.			AR Over Limit		\$1,096,600.00
24	EXC-1010	TRD-1004	MOV-1004	Uninvoiced Freight Accrual	High	M. Rossi	Aged Open	\$2,780,700.00	ACC-1005	Brent freight remains uninvoiced and is capitalized in landed inventory cost.			Open Vendor Reviews	1	

What this demonstrates

A single control surface brings together trade, movement, Incoterm, settlement, service-fee, PLA loss and credit/vendor review items. Open exceptions remain visible so operating issues are not hidden by workbook QA status.

6. Physical Execution and Demurrage Controls

	B	C	D	E	F	G	H	I	J
2	Demurrage Case ID	Movement ID	Trade ID	Asset ID	Asset Type	Asset Name	Current Asset Status	Port / Terminal	Counterparty
3	DMR-4001	MOV-1002	TRD-1002	VSL-4003	Vessel	MT Gulf Spirit	Available	Buckeye Bahamas	World Fuel Services
4	DMR-4002	MOV-1004	TRD-1004	VSL-4002	Vessel	MT Sea Pioneer	Available	Enterprise Houston	BP Energy
5	DMR-4003	MOV-1007	TRD-1007	VSL-4002	Vessel	MT Sea Pioneer	Available	Buckeye Bahamas	Shell Trading
6	DMR-4004	MOV-1009	TRD-1009	TRK-7001	Truck	Gulf Coast Trucking Co.	Available	Valero Port Arthur Truck Rack	Valero Marketing

	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y
2	NOR Tendered	NOR Accepted	Laytime Start	Laytime End	Laytime Allowed Hours	Laytime Used Hours	Excess Hours	Demurrage Rate Per Day	Calculated Demurrage	Variance	Responsible Party	Claim Status	Reason for Delay	Supporting Docs Complete	Notes
3	7/30/26 8:00	7/30/26 10:00	7/30/26 10:00	8/1/26 10:00	24.00	48.00	24.00	\$25,000.00	\$25,000.00	\$0.00	Receiving terminal / buyer side	Submitted	⌵ Berth congestion delayed discharge by one day.	Yes	⌵ Demurrage claim tracked separately from the commodity invoice.
4	8/8/26 6:00	8/8/26 8:00	8/8/26 8:00	8/10/26 8:00	24.00	48.00	24.00	\$35,000.00	\$35,000.00	\$0.00	Houston terminal / portfolio	Draft	⌵ Berth congestion at the nominated discharge terminal.	No	⌵ Signed statement of facts is still outstanding, so the case remains on hold.
5	8/23/26 8:00	8/23/26 10:00	8/23/26 10:00	8/25/26 10:00	24.00	48.00	24.00	\$30,000.00	\$30,000.00	\$0.00	Portfolio / charterer	Draft	⌵ Weather and berth congestion extended discharge by one day.	Yes	⌵ CFR seller-paid freight exposure; draft payable demurrage review.
6	8/26/26 9:00	8/26/26 9:00	8/26/26 9:00	8/26/26 14:00	2.00	5.00	3.00	\$2,280.00	\$285.00	\$0.00	Portfolio seller under DPU	Open	⌵ Three hours beyond the two-hour truck free-time allowance.	Yes	⌵ Truck detention rate of \$95/hour converted to a daily-equivalent rate for the shared control table.

Operational logic represented

Cases connect movement and trade IDs to assets, terminals and counterparties, then calculate laytime usage, excess hours, rate-per-day and demurrage exposure while preserving claim status and document completeness.

	A	B	C	D
1				
2				
3				
4	QA KPI	Value		Interpretation
5	Worksheets Reviewed	44		Includes this QA register.
6	Structured Tables Reviewed	105		All table ranges and over
7	Formula Errors	0		No spreadsheet formula-
8	Broken Named Ranges	0		Five legacy named-range
9	Hard Structural Issues Remaining	0		Primary keys, foreign key
10	Open Operational Exceptions	31		Business follow-up items
11	Overall QA Result	PASS		Workbook is structurally

Version 25 control refinement

CHGCTL-007 formalizes the invoice-balance re-performance: preserve the source-system reported outstanding balance and compare it with a calculated outstanding balance derived from Invoice Total and Payment Allocations.

Regression evidence

TEST-012 validates the synthetic exception cases, including active balance mismatches and a cancelled controlled exclusion. The companion Python application has 71 automated tests.

Interpretation

QA PASS means the workbook is structurally sound. Open operational exceptions are intentionally retained as business scenarios requiring review.

What this portfolio demonstrates

- Structured Excel table design, normalized IDs, primary/foreign-key thinking and controlled relationships.
- Payment allocation, invoice balance re-performance, exception classification, and separation of business state from control status.
- Physical-energy operations concepts including movements, inventory, service costs, demurrage, claims and month-end readiness.
- Governance through QA, change control, regression testing and a public Python companion for reproducible settlement controls.

Public companion repository github.com/uFiTech/energy-trade-controls